

# Case A: Next Auckland Scooter Project

## Overview



Ben Keith, a product manager at Next Auckland Company (Next Auckland), had just submitted a proposal for funding to the new product review committee. He recommended that Next Auckland introduce, through its retail distribution network, a new children's scooter based on a popular cartoon character. Ben felt that if he could launch by Christmas, Next Auckland could take advantage of what he felt would be strong market demand. However, due to the volatility of the market for these types of products, Ben could not be certain that the product would have the same level of market acceptance the following year. It is now **March 3, 2025**, and Ben knows he will have to manage the timing of the project carefully to launch the Scooter by **October 6, 2025** - just 31 weeks away.

Since Ben is an experienced project manager, he has established a preliminary list of activities for the Scooter Project, as shown in Table 2. However, he has sought advice from the teams who will do the tasks, and they have suggested the following changes to

the preliminary list of activities:

- Development Engineers commented on their preference to check the outcome of Product Design and Manufacturing Study at the same time before moving to Product Design selection. In other words, it would be ideal if those tasks finish at the same time, so they can check their outcomes on the same day. Assumer the time needed to check both outcomes is negligible (you don't need to add more days to their durations for 'outcome checks').

- The duration of the task "order components" takes 7 days. However, after ordering, shipping takes an additional 10 days.

- The duration of the task "order production equipment" takes 10 days. However, after ordering, shipping takes an additional 15 days.

Table 3 shows the personnel assigned to the Scooter project. However, concerned about the delivery, Ben has made some calls and managed to contract additional personnel to work on specific activities on the project. Since there is a staff shortage at Next Auckland, he requires the use of only additional resources that assist in meeting the deadline. The available personnel and impact on activity duration are presented in Table 4 (note: the additional resources cannot be assigned to other tasks).

Next Auckland is a relatively small company and uses a strong priority matrix to ensure they utilise their very limited workforce. The Project Priority Matrix for the Scooter Project is:

Table 1. Priority Matrix for Scooter Project.

|           | Time | Scope | Cost |
|-----------|------|-------|------|
| Constrain |      | x     |      |
| Enhance   | x    |       |      |
| Accept    |      |       | x    |

**Note:** The project team works eight-hour days, Monday through Friday, but not on public holidays.

The following public holidays are observed in New Zealand in 2025: New Year's Day (Jan 1st ), Day after New Year's Day (Jan 2nd), Waitangi Day (Feb 6th ), Good Friday (Apr 18th), Easter Monday (Apr 21st), Anzac Day (Apr 25th), King's Birthday (Jun 2nd), Matariki (Jun 20th), Labour Day (Oct 27th), Christmas Day (Dec 25th), Boxing Day (Dec 26th).

*Table 2. Activities needed for the new product introduction.*

| ID | Task Name                          | Duration | Predecessors | Resources                                                                  |
|----|------------------------------------|----------|--------------|----------------------------------------------------------------------------|
| 1  | <b>Scooter Development Project</b> |          |              |                                                                            |
| 2  | Market analysis                    | 10 days  |              | Marketing (4)                                                              |
| 3  | Product design                     | 15 days  | 2            | Marketing (1), Design (4), Development (2), Industrial (1), Purchasing (1) |
| 4  | Manufacturing study                | 15 days  | 2            | Industrial (4), Development (2)                                            |
| 5  | Product design selection           | 5 days   | 3,4          | Marketing (2), Design (3), Development (2), Industrial (2), Purchasing (1) |
| 6  | Detailed marketing plan            | 15 days  | 5            | Marketing (4)                                                              |
| 7  | Manufacturing process              | 30 days  | 5            | Design (1), Development (2), Industrial (4)                                |
| 8  | Detailed product design            | 20 days  | 5            | Marketing (2), Design (4), Development (2), Industrial (2), Purchasing (1) |
| 9  | Test prototype                     | 8 days   | 8            | Design (3), Development (2)                                                |
| 10 | Finalised product design           | 10 days  | 7,9          | Marketing (2), Design (3), Development (3), Industrial (2)                 |
| 11 | Order components                   | 7 days   | 10           | Purchasing (1)                                                             |
| 12 | Order production equipment         | 10 days  | 10           | Purchasing (1)                                                             |
| 13 | Install production equipment       | 20 days  | 11<br>12     | Development (3), Industrial (4), Design (1)                                |
| 14 | Celebrate                          | 1 day    | 6,13         | Development (4), Industrial (4), Design (4), Marketing (4), Purchasing (1) |

*Table 3. Scooter Project Resources*

| Resource             | Max. available | Std. Rate (\$/hr) | Cost/Use | Accrue   |
|----------------------|----------------|-------------------|----------|----------|
| Marketing specialist | 400%           | 65                | \$0.00   | Prorated |
| Design engineer      | 400%           | 70                | \$0.00   | Prorated |
| Development engineer | 400%           | 85                | \$0.00   | Prorated |
| Industrial engineer  | 400%           | 75                | \$0.00   | Prorated |
| Purchasing agent     | 100%           | 40                | \$0.00   | Prorated |

Table 4. Additional Resources

| Activity                     | Additional Resources            | Revised Duration Estimates |
|------------------------------|---------------------------------|----------------------------|
| Detailed marketing plan      | Marketing (2)                   | 7 days                     |
| Detailed product design      | Design (1), Development (1)     | 15 days                    |
| Install production equipment | Industrial (1), Development (1) | 10 days                    |
| Manufacturing process        | Industrial (2)                  | 20 days                    |

## Requirements

Ben needs to analyse the potential execution of this project carefully, and he seeks advice from students in the BUSINFO 710 course. You are asked to provide as much insight into the feasibility of executing this project with the information given.

You can use the following questions as guidelines to prepare a report to Ben.

- How long will it take to complete the project? Will the project be able to be completed by the October 6, 2025, deadline?
- What activities should Ben manage closely and avoid any delay?
- Is the initial plan feasible?
- Are there resources overallocated?
- Do you suggest a new feasible plan?
- Which additional resources would you choose to complete the project? You may need to explain your choices and the reasons for not choosing other options.
- How much is the project estimated to cost?
- What activity is estimated to cost the most to complete?
- It might be a good idea to show Ben the project's cash flow.

**Note:** You can skip some questions or add new ones. It is your decision as a consultant.

## Deliverables

- Prepare a business report with insight and recommendations to Ben. Show simple tables or graphs summarising relevant information, and include Gantt charts, Network Diagrams, Dashboards, etc, in Appendix. Max 2,500 words (including Appendix)
- Use the recommended report structure (adapt the names for headings and subheadings to communicate the contents of each section accordingly):
  - Cover page.
  - Executive Summary
  - Table of contents
  - Introduction
  - Problem and Objectives of the report
  - Analysis of the business scenario
  - Recommendations to the client
- Label any images and tables clearly and consistently in your report. *Referencing and citations are not required.*